Mining Silent Data Defects in LLM-Synthesized Agent Environments

Rahul Jha
Department of Computer Science
Texas Christian University
Fort Worth, Texas, USA
rahul.jha@tcu.edu

***Abstract*—This project tests whether tabular methods recognize SQL-valid injected value defects in unseen Agent World Model (AWM) environments. A reproduced initialization audit and a labeled mutation benchmark support separate analyses of execution failures and value anomalies. Eight methods are compared using disjoint training, validation, and test environments. Completed experiments examine corruption severity, omitted-type transfer, feature groups, repeated splits, and simulated low prevalence. Interpretation and an interactive demonstration use the same saved Random Forest. Results concern controlled mutations; they do not establish detection accuracy for naturally occurring semantic defects.**

I. Problem and Progress Since the Proposal

AWM synthesizes executable database-backed environments for training tool-using agents [1]. SQLite detects constraint violations but cannot determine arbitrary business meaning. The mining question is: *Can static statement features detect SQL-valid injected value defects in environments excluded from training?*

The approved proposal compared rules, Isolation Forest, and Random Forest. The expanded benchmark compares eight methods and tests sensitivity to corruption severity, excluded types, feature groups, and environment partitions. The dataset design also changed: unmodified statements are retained alongside separate mutations, instead of replacing a selected subset in place. One working instance is one unmodified or mutated INSERT statement; variants are dependent observations. Proposed positive-value changes, temporal ordering violations, and semantically incorrect references remain future work.

II. Data, Quality, and Leakage Checks

The public AgentWorldModel-1K dataset [2] is fully synthetic and released under CC-BY-4.0. The study uses its schema and sample-data JSONL files. The initialization audit reproduces the upstream SQL execution behavior [3]: 129,328 INSERT statements across 1,000 environments include 1,374 execution failures, affecting 217 environments; 1,054 failures are foreign-key violations. One environment has a malformed sample payload. These observations describe runtime defects, not labeled semantic errors.

Schemas mainly contain INTEGER, TEXT, DATETIME, and REAL attributes. Parsed source statements contain 45,982 explicit NULLs among 1,046,921 literals (4.39%) and 19,634 omitted column slots, which may receive defaults. There are 60,789 statements with dynamic date expressions, 31 unparsed statements, and 108 column/value mismatches. Only 3,931 of 18,465 table definitions (21.3%) contain CHECK clauses; constraint coverage is not universal.

The mutation builder preserves successful, supported source statements and checks each candidate under the original schema and preceding database state, with foreign keys enabled. A savepoint rolls back every trial before the next mutation. Label 0 means *unmodified executable source*, and label 1 means *accepted controlled mutation*; neither is a manually verified semantic label. The benchmark contains 619,619 instances in 999 environments: 127,954 unmodified examples and 491,665 accepted variants (79.35%). SQLite rejects 38,680 candidates. Date variants account for 75.3% of positives. The unmodified subset itself exceeds the 100,000-instance requirement.

Five mutation families use three severity levels. Money and quantity values become negative at different magnitudes; rating values become 99, 10, or 5.5; ages become 285, 150, or 120; date years become 1823, 1950, or the source year minus 30 (bounded below by 1900). Date shifts and age 120 can be plausible in some domains, so their labels represent the benchmark's assumptions. Unsupported or failed source statements, unchanged mutations, and SQLite-rejected candidates are counted separately in the build metadata. Source/mutated SQL, changed column, original/new values, and hashes are retained. An artifact audit reconciles RF metrics and replays SQL/features for 25 stratified pairs; their semantic judgments remain unreviewed.

The outer `GroupShuffleSplit` holds out 25% of environments, and a second grouped split reserves 20% of the remainder for validation (seed 42). Training uses 599 environments (376,201 rows), validation 150 (91,948), and test 250 (151,470). All statements and variants from an environment remain together. Assertions check disjoint environment sets. Parameters and scaling use training data; learned thresholds maximize validation F1; rules use a fixed Boolean cutoff. Test labels do not choose thresholds. Similar schemas may still occur across distinct environments, so grouping does not establish schema novelty.

TABLE I
HELD-OUT ENVIRONMENT RESULTS. LEARNED THRESHOLDS USE VALIDATION F1; RULES USE 0.5; CONSTANT-SCORE REFERENCE PREDICTS EVERY INSTANCE POSITIVE.

Method	AP	F1	Precision
Rules	0.8887	0.6723	0.9798
Mean/std Z-score	0.9341	0.8846	0.7931
Isolation Forest	0.8964	0.8862	0.7995
Local Outlier Factor	0.9509	0.8916	0.8402
Logistic regression	0.9977	0.9877	0.9905
Random Forest	0.9997	0.9947	0.9943
Histogram boosting	0.9992	0.9939	0.9946
XGBoost	0.9997	0.9956	0.9955
Constant reference	0.7931	0.8846	0.7931

III. FEATURES, METHODS, AND EVALUATION

The 26 pre-execution features comprise 11 literal summaries (signed-log numeric extrema, negative counts, parsed years, string lengths, and NULL counts), eight column/statement features (keyword indicators and column/value counts), and seven schema/position features (foreign keys, uniqueness, width, unknown columns, SQL length, and fill ratio). SQL date expressions are distinguished from literal dates in the quality analysis; a missing parsed year uses a fixed default, not evidence of a true year.

The rule baseline flags row-level numeric/date patterns in the presence of relevant column keywords. A second baseline takes the maximum absolute mean/std Z-score across features. Isolation Forest (200 trees) [4] and novelty-mode LOF ($k = 20$) [5] fit unmodified training examples. LOF uses training-only standardization and a deterministic cap of 25,000 fitting examples. Supervised methods are scaled, class-balanced logistic regression; Random Forest (200 trees) [6]; histogram gradient boosting (maximum 300 iterations); and XGBoost (300 trees) [7], with a training-derived positive-class weight. Full configurations are saved with the results.

The primary metric is average precision (AP), computed by `average_precision_score`; it is not trapezoidal PR-AUC. Random-ranking AP equals positive prevalence. F1 uses validation-selected thresholds (rules: fixed 0.5), and Precision@100/Recall@100 measure ranking. All main results in Table I concern the same held-out environments.

IV. COMPLETED EXPERIMENTS AND FINDINGS

The test set has 120,137 positives and 31,333 negatives (baseline AP 0.7931). XGBoost has AP 0.9997 and F1 0.9956; differences among ensembles are small. Z-score's validation-selected cutoff predicts every test row positive, giving the same F1 (0.8846) as the constant reference despite better ranking. This shows why F1 alone is insufficient at this prevalence.

Severity: Each evaluation subset contains all unmodified test negatives plus mutations at one severity; reported positive counts are not total test counts. Each subset has 31,333 negatives and 40,043/40,051/40,043 positives (obvious/medium/subtle), with baseline AP about 0.5610. Isolation Forest AP is 0.7771/0.7612/0.7207; histogram boosting gives 0.9983/0.9982/0.9977.

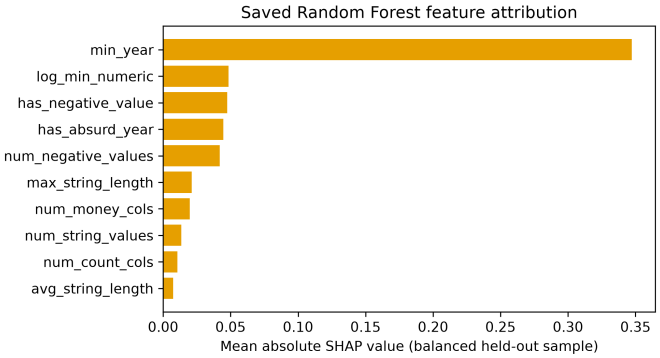


Fig. 1. Mean absolute SHAP values for the saved Random Forest on the balanced test sample. These explain model associations, not causal defect validity.

Omitted-type generalization: A type is removed from training and validation, then evaluated with unmodified test negatives. Different supports imply different random-ranking baselines. Omitted dates give AP 0.7410, below the 0.7454 prevalence baseline (91,734 positives), providing no ranking-transfer evidence. Ratings give AP 0.0944 versus baseline 0.0385 (1,253 positives); ages give 0.0053 versus 0.0032 (102 positives). Numeric-type transfer is stronger: money/inventory AP is 0.8861/0.8818 versus baselines 0.3749/0.2085.

Feature ablation: The same partitions compare 11, 19, and 26 cumulative features with histogram boosting. AP is 0.9992, 0.9997, 0.9992; F1 is 0.9924, 0.9955, 0.9939, respectively. The observed differences do not establish that a feature group is necessary or causally responsible for detection.

Repeated splits and prevalence: Seeds 42, 43, and 44 repeat the grouped evaluation for logistic regression, histogram boosting, and Isolation Forest. AP mean $\pm$ split standard deviation is 0.99728 ± 0.00035 for logistic regression; 0.99951 ± 0.00029 for histogram boosting; 0.89837 ± 0.00497 for Isolation Forest. Simulated 1% and 5% positive subsets reuse fitted models and fixed validation thresholds; this is sensitivity analysis rather than an estimate of natural defect prevalence. At 1% (316 positives, 31,333 negatives), histogram boosting AP is 0.9437, but precision is 0.3261 and F1 0.4914. At 5% (1,649 positives, the same negatives), AP is 0.9857 and precision 0.7157. Fixed-threshold alert quality changes substantially.

V. INTERPRETATION AND REPRODUCIBLE DEMONSTRATION

The saved 200-tree Random Forest supplies the reported RF predictions, SHAP [8], and Streamlit demonstration. SHAP uses 400 test examples (200 per label, seed 42), so global attribution summaries reflect balanced sampling rather than deployment prevalence. Four workers compute exact TreeExplainer values without retraining. The leading mean absolute attributions are `min_year` (0.3474), `log_min_numeric` (0.0486), `has_negative_value` (0.0476). This reflects the controlled date/numeric mutations.

RF produces 691 false positives (2.21% of unmodified test examples) and 594 false negatives (0.49% of variants). A points redemption of -1500 in `hr_employee_management_1` receives score 1.0 despite being an unmodified debit. In `ats_applicant_tracking_system_3`, changing a rating from 4 to 10 receives score 0.005, below the RF threshold 0.2871; the row's identifier 15 leaves its maximum numeric feature unchanged. Exact original and mutated statements are preserved in the error catalog. Row-wide extrema can mask an altered rating behind a larger identifier; a domain-valid negative balance can resemble an injected monetary defect. These are hypotheses supported by inspected examples, not a complete error taxonomy.

The demonstration accepts edited SQL, displays the shared detector's score, and independently tests SQLite acceptance under the actual source schema/state. W&B records configurations, metrics, tables, and plots; local offline evidence and access requirements are documented for final-project bonus evaluation. Weekly Git activity, interpretation, and a runnable demonstration address the other noncompetition bonus items, without claiming credit already awarded.

VI. Limitations and Remaining Work

The benchmark is dominated by repeated date/value mutations, and unmodified rows may already contain natural defects. Mutation-family labels are lexical recipes: matching `count` also selects the Boolean column `is_bank_account` as an inventory-family candidate. Keyword eligibility does not verify business rules. High AP therefore indicates recognition of these transformations under the stated protocol. Three seeds provide a limited stability check, not a confidence interval. Examining test errors informs further work, so a new final holdout or explicit exploratory designation is needed for conclusions after feature revisions.

During October, representative source/mutated pairs will be reviewed across domains and column-aligned features designed. November experiments will test those features and plausible positive-value or temporal-consistency errors. Configuration will be frozen before evaluation; revisions will be labeled exploratory if an untouched holdout is unavailable. Final bonus evidence must include actual tracking runs, accessible code, continued weekly commits, and a demonstrated end-to-end example; September activity is not established by the current Git history.

Working code and reproduction: github.com/jharahul968/awm-silent-defects. The README documents data access, module commands, dependencies, tests, saved artifacts, and the demo. The submission snapshot must include the reviewed revision; the repository's upstream code and assistance attribution are documented separately.

References

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